

Sitan Yan

EDUCATION

Huazhong University of Science and Technology (HUST)

M.Eng. in **Mechanical Engineering** (*Expected*)

- GPA: 3.95/4.0

Wuhan, China

Sep 2024-Jun 2027

B.Eng. in **Mechanical Design, Manufacturing and Automation**

- GPA: 3.96/4.0

- Selected Awards: Distinguished Graduate of HUST (2024)

Sep 2020-Jun 2024

PUBLICATIONS

S. Yan, W. Xu, L. Zeng, and W. Li, "Reliability-aware shape-from-focus for reflective 3-D micro-hole measurement," manuscript under major revision at *IEEE Trans. Instrum. Meas.*, 2026.

T. Chen, J. Han, J. Wang, **S. Yan**, C. Wan and F. Tian, "Preliminary Design and Prototype Development of an Air-ground Carrier Platform," 2023 International Conference on Unmanned Aircraft Systems (ICUAS), Warsaw, Poland, 2023, pp. 235-240, doi: 10.1109/ICUAS57906.2023.10156435.

PATENTS

F. Tian, **S. Yan**, J. Wang, T. Chen, J. Wan, and J. Han, "A deformable amphibious robot for open space," Chinese Patent CN219191842U, approved on Jun. 16, 2023. [Utility Model]

F. Tian, T. Chen, **S. Yan**, J. Wang, J. Wan, and J. Han, "A quadcopter driven flying car," Chinese Patent App. CN202211663206.3, filed on Dec. 23, 2022. [Invention]

RESEARCH INTERESTS

Robotic Perception and Control, Automated Optical Metrology, Motion Planning and Optimization, SFF Reconstruction

RESEARCH

Prof. Christian Claudel's Research Group, UT-Austin

Austin, TX (Remote)

1. **Robotic Dynamic Obstacle Avoidance Control based on Partial RGB-D Observations**

Jan 2026-Present

- Implemented and evaluated a closed-loop obstacle-avoidance framework in ROS 2/Gazebo, using RGB-D reconstruction and local point-cloud processing to provide online obstacle geometry to robot AUBO i16H in static and dynamic scenes.
- Estimated dynamic-obstacle occupancy by extracting local point clouds and initializing bounding ellipsoids with PCA and minimum-volume enclosing ellipsoid (MVEE) optimization.
- Trained a residual network with a joint coverage-compactness loss to refine obstacle bounding ellipsoids under camera occlusion and depth noise.
- Combined Kalman-filter trajectory prediction with MPC-based spatial constraints to incorporate moving-obstacle estimates into robotic-arm motion planning.

2. **Focus-Constrained Visual Predictive Control for Robotic Micro-Hole Localization**

Jun 2026-Aug 2026

- Developed a robot-assisted microscopic measurement framework for local micro-hole alignment under limited field of view and shallow depth of field.
- Modeled coordinated motion of the robot and linear stage to support target positioning and fine optical adjustment.
- Investigated focus-aware visual feedback for local search and centering under actuator and workpiece-clearance constraints.
- Integrated the experimental measurement workflow and prepared system demonstrations; detailed methods and quantitative evaluation are reserved for publication.

State Key Laboratory of Digital Manufacturing Equipment and Technology

Wuhan, China

Advisor: Prof. Wenlong Li & Prof. Bo Tao

1. **Reliability-Aware Microscopic 3D Reconstruction System for Reflective Micro-Holes**

Jun 2025-Present

- Designed and integrated a microscopic shape-from-focus (SFF) metrology system with 10X lenses, polarizers, and translation stages to acquire focus stacks of reflective industrial micro-holes.
- Developed image-processing methods to improve microscopic observations of reflective micro-hole surfaces under challenging imaging conditions.
- Implemented a reliability-aware reconstruction framework to improve surface recovery from microscopic image sequences.
- Evaluated reconstruction quality against an independent reference measurement system; detailed algorithms and quantitative comparisons are reserved for publication.

2. **Robotic Trajectory Planning for Aircraft CFRP Horizontal Stabilizer Rib Machining**

Sep 2024-May 2025

- Derived ABB-robot kinematics using the modified Denavit-Hartenberg convention and constructed a Jacobian-based static stiffness model; mapped workspace stiffness through Monte Carlo sampling to support machining-pose selection.
- Developed a time-optimal trajectory planner in MATLAB using fifth-order polynomial interpolation and particle swarm optimization with dynamic inertia weights, reducing predicted machining time by 40.8% relative to GA.
- Generated stiffness and force-ellipsoid maps through Monte Carlo sampling and used them to identify candidate high-stiffness pose regions for robotic machining.
- Designed and fabricated a split-type dust-control fixture using SolidWorks and 3D printing to improve sealing around the

high-speed spindle.

- Validated trajectories in RobotStudio and conducted CFRP orthogonal-cutting experiments; optimized cutting parameters to reduce surface roughness by 13.6% to below 8.2 μm .

3. R&D of High-Performance PMSM Servo Drive System

Sep 2023-Mar 2024

- Designed GD32F470-based servo-drive control and power boards with RS-232/RS-485 communication, optocoupler isolation, dynamic braking, and fault-protection circuits.
- Implemented a seven-segment SVPWM model in Simulink and experimentally identified motor resistance and inductance using voltage and LR step-response tests.

Air-Ground Carrier Platform Based on a Lead Screw-Linkage Synergetic Morphing Mechanism

Jun 2022-Jun 2023

Advisor: Prof. Fengnian Tian, HUST

- Co-designed a lead-screw/linkage morphing mechanism that synchronously actuates four arms and integrated the mechanism with the drone landing-gear and ground-chassis architecture.
- Modeled the morphing mechanism kinematics and evaluated rigidity and self-locking through finite-element analysis, revising the geometry to reduce stress concentration and vibration risk.
- Fabricated and tested a physical prototype, diagnosed mechanical jamming, and evaluated configuration switching through ground and field flight tests.

INTERNSHIP

AVIC Beijing Aeronautical Precision Machinery Institute

Beijing, China

Project: 5-Axis Measurement System for Aero-engine Turbine Blade Cooling Holes

Jul 2025-Present

- Developed a Python/PyQt5 MVC application integrating OpenCASCADE for CAD visualization and PyVista for point-cloud processing.
- Implemented B-Rep topology and ray-casting algorithms to extract cooling-hole center coordinates and axis vectors automatically from STEP models.
- Formulated multi-hole measurement sequencing as an extended multi-TSP and combined particle swarm optimization with 2-opt local search, reducing travel distance by 27.68% relative to PSO.
- Generated five-axis position and orientation trajectories using quaternion-based orientation handling and cubic B-spline interpolation with C2 continuity, reducing gimbal-lock and resonance-prone motions.
- Implemented synchronized control of three-axis linear stages, piezoelectric stages, and two-axis rotary tables through the I++ DME protocol, TUCAM SDK, and socket communication.

SELECTED COURSE PROJECTS

TwinCAT-based Open SCARA Robot Motion Control

Mar 2025-May 2025

- Derived forward and inverse kinematics for a 4-axis (3R1P) SCARA robot, mapped the virtual NC axes to physical servo axes, and configured EtherCAT communication.
- Implemented a reusable FB_AxisFun function block integrating axis enabling, jogging, and absolute-position control for the multi-axis TwinCAT program.
- Developed an HMI with Python, PySide2, and PyADS, and integrated NCI interpolation with G-code parsing to achieve real-time bidirectional communication with TwinCAT and validate trajectory execution

Rolling Bearing Fault Diagnosis Based on Vibration Signal Analysis

Oct 2022-Dec 2022

- Built a vibration-measurement setup with a Baumer inductive sensor and implemented FFT-based analysis in MATLAB to identify inner-race, outer-race, and rolling-element faults in SKF 6205 bearings with 0.1778 mm and 0.5334 mm defects.

HONORS & AWARDS

First Prize, Chinese Mathematics Competitions (CMC)

Nov 2022

Third Prize, China Collegiate Intelligent Robotics Creative Competition

Nov 2022

Academic Excellence Scholarship

Jul 2021

Self-Improvement Scholarship

Jul 2021

Second Prize, National College Students Advanced Drawing Technology and Product Information Modeling Innovation Competition

Jul 2021

Second Prize, Hubei Provincial College Students Advanced Drawing Technology and Product Information Modeling Innovation Competition

May 2021

Outstanding Freshman Scholarship

Mar 2021

Freshman Self-Improvement Scholarship

Mar 2021

SKILLS

Programming: C++, Python, MATLAB

Tools: ROS/ROS2, Gazebo, MoveIt, SolidWorks, Simulink

Certifications: National Computer Rank Examination (NCRE) Level 2 in C++; NCRE Level 3 in Database Technology

Leadership: Vice Director, Social Practice Department, HUST (2021-2022)